

HW2 prac.
Con.

* Sums to
Remember:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(2n+1)(n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$$

$$\sum_{j=0}^m \lfloor \sqrt[3]{j} \rfloor$$

$$(k+1)^3 = k^3 + 3k^2 + 3k + 1$$

$$\sum_{k=1}^n k(3k^2 + 3k + 1) = 3 \sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + \sum_{k=1}^n k$$

reating
prob.
like an
edge case

$$\sum_{j=0}^m \lfloor \sqrt[3]{j} \rfloor = \sum_{j=0}^{q^3-1} \lfloor \sqrt[3]{j} \rfloor + \sum_{j=q^3}^m \lfloor \sqrt[3]{j} \rfloor$$

$$= 6(q-1) + \sum_{q^3}^m q$$

$$q = \lfloor \sqrt[3]{m} \rfloor$$

$$= 6(q-1) + q(m - q^3 + 1)$$

$$q^3 \leq m$$

↙ closest cube to m

↳ simplify, leave answer with m, q

m is the

upper limit of

our sum (could be any #)